Muhammed Roshan M K

Software Engineer | Python | FastAPI | Backend Development
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PROFESSIONAL SUMMARY

Backend-focused Software Engineer with 2.5+ years building production-grade, cloud-native systems in Python for regulated healthcare products. Designs scalable async REST APIs and modular microservices with FastAPI, SQLAlchemy, and Pydantic, backed by PostgreSQL, Redis caching, and Kafka event flows. Recently extended into patient-facing mobile backends: AI RAG chatbot orchestration, self-hosted Chatwoot on GCP Compute Engine, Firebase single-device login, and scheduled push notifications. Owns work end to end (architecture, data modelling, JWT/RBAC security, Docker, CI/CD, and GCP deployment) and collaborates across mobile, frontend, QA, and DevOps in Agile teams.

TECHNICAL SKILLS

Languages: Python (primary), SQL, JavaScript, Java

Backend: FastAPI, REST API Design, API Versioning, Microservices, Async/Await, Dependency Injection, Middleware, Background Tasks, Pagination & Filtering, Pydantic, OpenAPI/Swagger

AI & Chat: RAG chatbot integration, LLM service orchestration, Chatwoot (self-hosted), conversational check-in flows

Databases: PostgreSQL, MySQL, SQLite, SQLAlchemy ORM, Alembic Migrations, Schema Design, Indexing, Query Optimization

Caching & Messaging: Redis (caching, invalidation), Apache Kafka (producers, consumers, event-driven architecture)

Mobile Backend: Firebase Cloud Messaging (push notifications), Firebase Auth, device-token session management

Cloud & DevOps: Google Cloud Platform (Compute Engine, Cloud Storage, Artifact Registry), AWS (basic), Docker, Docker Compose, CI/CD (Bitbucket Pipelines, YAML), Poetry

Practices: Repository & Service Layer Patterns, JWT Authentication, RBAC, Structured Logging, Exception Handling, Environment Management, Git Workflows, Code Reviews, Agile/Scrum, Technical Documentation

PROFESSIONAL EXPERIENCE

Software Engineer, Gadgeon Smart Systems Pvt. Ltd.

Kochi, Kerala | Jan 2024 - Present

PregPath App | Patient Mobile App with AI RAG Chatbot | Backend Developer

- Built the backend for the patient-facing mobile companion to the PregPath clinical web platform, extending care from doctors, nurses, and staff directly to expectant mothers.
- Provisioned and configured a GCP Compute Engine VM hosting self-hosted Chatwoot with integrated Redis and PostgreSQL, giving the team full control over patient conversation data.
- Developed and deployed a chat orchestrator middleware service on Google Cloud that routes messages between Chatwoot and the AI RAG chatbot, enabling context-aware answers to patient questions.
- Designed weekly AI-driven check-in flows where the chatbot gathers patient status updates, surfacing health signals to the clinical team.
- Implemented single-device login using Firebase device tokens, automatically invalidating sessions on other devices to protect patient health data.
- Set up Firebase Cloud Messaging push notifications for scheduled weekly check-in reminders, driving consistent patient engagement.
- Built FastAPI endpoints for the app's dashboard, downloads, and notification screens, with paginated responses and strict Pydantic validation.

PregPath AI | AI-Driven Maternal Healthcare Platform | FastAPI Backend Developer

- Designed and shipped scalable async REST APIs in FastAPI powering real-time maternal health monitoring, risk assessment, and the end-to-end patient journey, with versioned endpoints and strict Pydantic validation on clinical payloads.
- Architected the backend as modular microservices (core API, rule engine, dashboard API) with clear service boundaries and shared contracts, enabling independent development and deployment.

- Implemented Redis caching with targeted TTL and invalidation strategies for patient and reference data, improving response times on read-heavy dashboard endpoints.
- Built Kafka producers and consumers for event-driven communication between services, decoupling long-running risk assessment workloads from the request path.
- Modelled and evolved the PostgreSQL schema with SQLAlchemy and Alembic for safe production releases; tuned indexing and resolved N+1 query patterns on high-volume clinical tables.
- Packaged the shared data-access layer into a versioned Python module distributed via Poetry, eliminating duplicated repository code across three services.
- Automated build and release through YAML Bitbucket Pipelines that publish the package to Google Artifact Registry on merge to develop.
- Secured protected health data with JWT authentication and role-based access control, enforced through reusable FastAPI dependencies and middleware.

Medical AI | AI-Driven General Healthcare Platform | FastAPI Backend Developer

- Built end-to-end FastAPI services for patient registration, diagnosis monitoring, treatment tracking, and recovery management, with paginated, filterable endpoints and auto-generated OpenAPI docs.
- Designed the PostgreSQL data model and tuned indexing and query performance to keep clinical read/write paths reliable under production load.
- Developed an automated intervention-report service that extracts patient data, renders clinical PDFs as a background task, and uploads them to Google Cloud Storage.
- Resolved a critical reporting defect by migrating PDF generation from FPDF to ReportLab with custom paragraph-breaking logic, eliminating layout and pagination failures on long reports.
- Containerized services with Docker and Docker Compose, managing configuration and secrets through environment-based settings across dev and production.
- Instrumented structured logging and centralized exception handling, and drove root-cause analysis on live issues alongside QA and DevOps.
- Reviewed peer pull requests and documented API contracts and service behaviour in Confluence.

EDUCATION

B.Tech in Computer Science, LBS College of Engineering

Kasaragod, Kerala | Jul 2019 - Jun 2023